Please check the examination details below before entering your candidate information

| Candidate surname | Other names |
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**Pearson Edexcel International Advanced Level**

Centre Number | Candidate Number

# Wednesday 8 January 2020

Morning (Time: 1 hour 30 minutes) | Paper Reference **WMA11/01**

# Mathematics

## International Advanced Subsidiary/Advanced Level
## Pure Mathematics P1

**You must have:**
Mathematical Formulae and Statistical Tables (Lilac), calculator

Total Marks

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

## Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

## Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 11 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

## Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

*Turn over* ▶


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**1.** Find, in simplest form,

$$\int \left( \frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) \mathrm{d}x$$

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**Question 1 continued**

**Q1**

**(Total 4 marks)**

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**2.** Given $y = 3^x$, express each of the following in terms of $y$. Write each expression in its simplest form.

(a) $3^{3x}$

**(1)**

(b) $\dfrac{1}{3^{x-2}}$

**(2)**

(c) $\dfrac{81}{9^{2-3x}}$

**(2)**

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**Question 2 continued**

**Q2**

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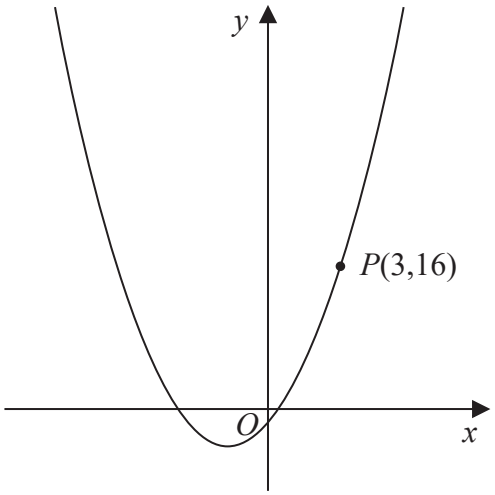


**Figure 1**

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at $P$.

**(2)**

The point $Q$ with $x$ coordinate $3 + h$ also lies on the curve.

(b) Find, in terms of $h$, the gradient of the line $PQ$. Write your answer in simplest form.

**(3)**

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

**(1)**

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**Question 3 continued**

**Q3**

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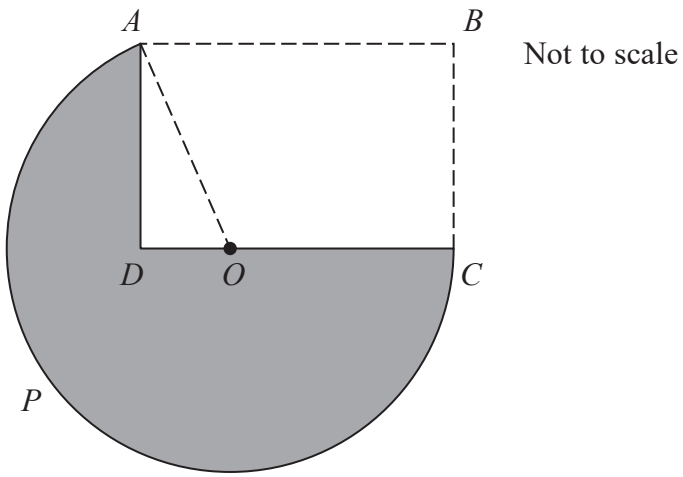


**Figure 2**

Figure 2 shows the plan view of a house *ABCD* and a lawn *APCDA*.

*ABCD* is a rectangle with $AB = 16\,\text{m}$.

*APCOA* is a sector of a circle centre *O* with radius 12 m.

The point *O* lies on the line *DC*, as shown in Figure 2.

(a) Show that the size of angle *AOD* is 1.231 radians to 3 decimal places.

**(2)**

The lawn *APCDA* is shown shaded in Figure 2.

(b) Find the area of the lawn, in $\text{m}^2$, to one decimal place.

**(4)**

(c) Find the perimeter of the lawn, in metres, to one decimal place.

**(3)**

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**Question 4 continued**

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**Question 4 continued**

**Q4**

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**5.** (a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0$$

**(3)**

(b) Hence find all real solutions of

$$20(y + 3)^{\frac{3}{2}} - 50(y + 3) - 30(y + 3)^{\frac{1}{2}} = 0$$

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**Question 5 continued**

**Q5**

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**6.** The line $l_1$ has equation $3x - 4y + 20 = 0$

The line $l_2$ cuts the $x$-axis at $R(8,0)$ and is parallel to $l_1$

(a) Find the equation of $l_2$, writing your answer in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers to be found.

**(3)**

The line $l_1$ cuts the $x$-axis at $P$ and the $y$-axis at $Q$.

Given that $PQRS$ is a parallelogram, find

(b) the area of $PQRS$,

**(3)**

(c) the coordinates of $S$.

**(2)**

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**Question 6 continued**

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**Question 6 continued**

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**Question 6 continued**

**Q6**

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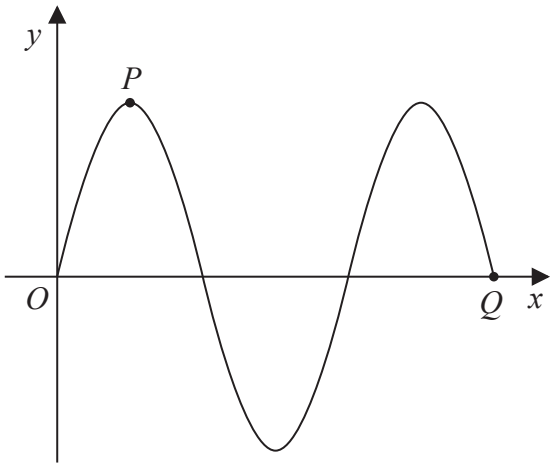


**Figure 3**

Figure 3 shows part of the curve $C_1$ with equation $y = 3\sin x$, where $x$ is measured in degrees.

The point $P$ and the point $Q$ lie on $C_1$ and are shown in Figure 3.

(a) State

(i) the coordinates of $P$,

(ii) the coordinates of $Q$.

**(3)**

A different curve $C_2$ has equation $y = 3\sin x + k$, where $k$ is a constant.

The curve $C_2$ has a maximum $y$ value of 10

The point $R$ is the minimum point on $C_2$ with the smallest positive $x$ coordinate.

(b) State the coordinates of $R$.

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**Question 7 continued**

**(Total 5 marks)**

Q7

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**8.** The straight line $l$ has equation $y = k(2x - 1)$, where $k$ is a constant.

The curve $C$ has equation $y = x^2 + 2x + 11$

Find the set of values of $k$ for which $l$ does not cross or touch $C$.

**(6)**

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**Question 8 continued**

**Q8**

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**9.** **In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \qquad x > 0$$

Find the $x$ coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

**(6)**

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**Question 9 continued**

**Q9**

**(Total 6 marks)**

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**10.** The curve $C_1$ has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (4x - 3)(x - 5)^2$$

(a) Sketch $C_1$ showing the coordinates of any point where the curve touches or crosses the coordinate axes.

**(3)**

(b) Hence or otherwise

(i) find the values of $x$ for which $\mathrm{f}\left(\frac{1}{4}x\right) = 0$

(ii) find the value of the constant $p$ such that the curve with equation $y = \mathrm{f}(x) + p$ passes through the origin.

**(2)**

A second curve $C_2$ has equation $y = \mathrm{g}(x)$, where $\mathrm{g}(x) = \mathrm{f}(x + 1)$

(c) (i) Find, in simplest form, $\mathrm{g}(x)$. You may leave your answer in a factorised form.

(ii) Hence, or otherwise, find the $y$ intercept of curve $C_2$

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**Question 10 continued**

**Q10**

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**11.** A curve has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}''(x) = \frac{6}{\sqrt{x^3}} + x \qquad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $\mathrm{f}'(x) = -4$ at $P$,

(a) find the equation of the normal at $P$, writing your answer in the form $y = mx + c$, where $m$ and $c$ are constants,

**(3)**

(b) find $\mathrm{f}(x)$.

**(8)**

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**Question 11 continued**

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**Question 11 continued**

**Q11**

**(Total 11 marks)**

**TOTAL FOR PAPER IS 75 MARKS**

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